

Caden Phillips

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MPhys Astrophysics student at the University of Liverpool, specialising in compact objects, computational modelling, and time-domain astrophysics. Research experience in Fisher matrix forecasting, TOV solvers, hybrid equation-of-state construction, X-ray timing analysis of NICER data and spectral fitting of XMM-Newton observations. Experienced in building end-to-end analysis workflows, from numerical solvers to data reduction, with strong Python and scientific computing skills. Seeking a PhD in X-ray studies of accreting compact objects.

EDUCATION

MPhys Astrophysics, University of Liverpool

Expected 2027

Projected First Class · Year 3: 73.5%

Academic Advisor: Dr. Christopher Copperwheat

Selected modules: Numerical Methods in Physics, Relativity and Cosmology, Advanced Observational Astrophysics, The Interstellar Medium, Physics of Planets

Full transcript available upon request.

RESEARCH EXPERIENCE

Independent Study — X-ray Weak Active Galactic Nuclei

Summer 2026

Self-directed project based on an internship offered by LEAPS Leiden ESA

- Built an end-to-end Python pipeline automating XMM-Newton EPIC data reduction (SAS), spectral fitting (PyXSPEC) and Bayesian hardness-ratio estimation (BEHR), applied to 90+ quasars at redshift 1.8 to 2.2 drawn from a crossmatch of the 5XMM-DR15 and SDSS DR16Q catalogues.
- Classified sources as X-ray weak, normal or strong via $\Delta\alpha_{\text{ox}}$, and examined how photon index, hardness ratio and fractional variability vary across the $\Delta\alpha_{\text{ox}}$ distribution, accounting for count-dependent fit reliability and sample selection bias.
- Validated pipeline using well-studied sources and compared outputs with published data.
- Compared target outputs against 5XMM-DR15 catalogue.
- Found a weak-to-no correlation between photon index, Γ , and $\Delta\alpha_{\text{ox}}$.

Independent Study — HEASARC/NICER Analysis Pipeline

Winter 2025 – Spring 2026

Self-directed observational analysis project

- Developed a Python pipeline integrating HEASoft tools via Bash to automatically download and organise archival NICER X-ray observations from the HEASARC database.
- Automated event filtering and calibration with scripted XSELECT workflows before light-curve extraction, across up to hundreds of ObsIDs.
- Applied to archival observations of transitional millisecond pulsars.
- Produced calibrated X-ray light curves and diagnostic plots to enable time-domain comparison of source variability across observing epochs.

Internship — Institute of Nuclear Physics (IFJ PAN), Kraków, Poland *Jul 2025 – Aug 2025*
Supervisor: Dr. David Álvarez Castillo

- Applied a Fisher Information Matrix framework to estimate tidal deformability uncertainties and assess error propagation in binary neutron star systems.
- Developed and implemented a Tolman–Oppenheimer–Volkoff (TOV) solver to compute mass-radius and tidal deformability relations for 62 hybrid equations of state.
- Processed 62 hybrid equations of state, generating mass-radius and tidal deformability curves (approximately 400 stellar configurations per EoS) and analysing stable and unstable branches.
- Compared model predictions with GW170817 50% and 90% credible regions; none of the 62 hybrid EoS fell within either.
- Co-presented results at the IFJ PAN Summer Internship Symposium 2025 with a fellow intern.

Field Work — Teide Observatory, Tenerife, Spain
Supervisor: Dr. Daniel Harman

Jun 2025

- Operated the 0.8 m IAC-80 telescope and a Celestron C14 Edge HD system with a QHY-163M/C CMOS detector.
- Identified and observed targets as a team, including M7, M13 and Abell 39.
- Calibrated and reduced multi-band CMOS data to construct Hertzsprung–Russell diagrams and composite images.

RESEARCH OUTPUTS

- **Report & presentation:** *Physical Inputs for Simulations of Gravitational Wave Emissions from Compact Stars within the Einstein Telescope Detection Range*. Institute of Nuclear Physics (IFJ PAN), Kraków, 2025; presented at the IFJ PAN Summer Internship Symposium 2025. [PDF]. [Slides]. [GitHub].
- **Report:** *A study of X-ray properties of Active Galactic Nuclei at $z = 1.8\text{--}2.2$ across the $\Delta\alpha_{\text{ox}}$ spectrum*. Independent Study, 2026. [PDF]. [GitHub].
- **Code:** *Using HEASoft to Analyse Archived NICER Data of tMSPs*. Independent Study, 2026. [GitHub].

TECHNICAL SKILLS

- **Programming:** Python (NumPy, SciPy, Matplotlib, Astropy), C++, Bash.
- **Tools & software:** HEASoft, XMM-Newton SAS, XSPEC/PyXSPEC, BEHR, RNS, LORENE, ROOT, Jupyter, Git, L^AT_EX.
- **Scientific methods:** TOV and hybrid EoS modelling, Fisher matrix forecasting, spectral fitting, light-curve analysis, numerical ODE solvers.

AWARDS & MEMBERSHIPS

- Awarded competitive summer internship placement at IFJ PAN (Poland) through the PHYS309 Internship Module. [PDF].
- Associate Member, Institute of Physics (IOP) *since 2023*.

INTERESTS

Astronomy and astrophotography, hiking, cycling, history.